

QIRAN JIA

BIostatistician · Computational Biology & Machine Learning Scientist

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SUMMARY

Biostatistician and bioinformatician developing novel statistical and machine learning methods, software, and analytical pipelines for complex, high-dimensional biomedical data, with expertise in multimodal data integration, statistical inference, and deep learning across bulk and single-cell multi-omics, longitudinal cell imaging, and 3D medical imaging

EDUCATION

University of Southern California

Ph.D. in Biostatistics

Los Angeles, CA

Dec. 2026 (expected)

New York University

New York, NY

M.S. in Biostatistics

May 2022

University of California, Santa Barbara

Santa Barbara, CA

B.A. in Statistics and Chemistry

Dec. 2019

EXPERIENCE

Cellanome

San Diego, CA

COMPUTATIONAL BIOLOGY INTERN

Jun. 2026 - Aug. 2026

- Analyzed multimodal single-cell datasets generated on the R3200 platform with CellCage™ technology across diverse experimental settings using rigorous, streamlined, and reproducible workflows to uncover biological insights
- Developed a deep-learning trajectory modeling framework and supporting internal software for analyzing longitudinal live-cell imaging, capturing cellular morphodynamic patterns and revealing novel biological cell states beyond snapshot morphology or endpoint transcriptomics alone
- Applied novel visualization tools, integrative morphodynamic–transcriptomic analyses, and pathway enrichment analysis to characterize dynamic cell states and their associated transcriptomic profiles

University of Southern California

Los Angeles, CA

GRADUATE RESEARCH ASSISTANT, BIOSTATISTICS

Aug. 2022 - Present

- Engineered LUCID, a statistical quasi-mediation framework and R package for multi-omics clustering and inference along with high-dimensional exposures and outcomes, achieving 120× faster computation while extending functionalities including variable selection, uncertainty quantification, missing-data handling, flexible integration strategies, and visualization, with application to several environmental health cohort studies
- Developed Praxis-BGM, a scalable Bayesian transfer-learning framework for Gaussian mixture models (GMMs) leveraging natural-gradient variational inference and informative priors to improve stability, accuracy, and interpretability in high-dimensional, small-sample molecular cohort studies, with applications to TCGA and METABRIC data
- Collaborated regularly with biologists, clinicians, and epidemiologists on method development and applications across environmental health, multimodal single-cell omics, and spatial transcriptomics, providing statistical and computational consulting

New York University

New York, NY

GRADUATE RESEARCH ASSISTANT, BIOSTATISTICS

Feb. 2021 - Sep. 2022

- Designed BiTr-UNet, a CNN–Transformer architecture leveraging self-attention and advanced post-processing techniques for 3D brain tumor segmentation, integrating local and global representations and substantially improving segmentation performance
- Co-developed scalable Hidden Markov Random Field (HMRF)- and deep-learning-based spatial multiple-testing frameworks for inference across hundreds of thousands of FDG-PET voxels, improving power while controlling false discovery rate (FDR)

SAMPLE PROJECTS

Praxis-BGM — Bayesian semi-supervised transfer learning for prior-augmented GMM clustering of bulk and single-cell omics using natural-gradient variational inference [\[repo\]](#)

LUCIDus — R package for integrative clustering of multi-omics data with high-dimensional exposures and outcomes, association inference, missing-data handling, variable selection, and visualization [\[repo\]](#)

BiTr-UNet — CNN–Transformer network for 3D MRI brain tumor segmentation, combining local convolutional features with global transformer representations [\[repo\]](#)

DeepFDR — Deep-learning framework for spatial multiple testing and FDR control in large-scale 3D neuroimaging [\[repo\]](#)

SKILLS

Programming Python, R; JAX, PyTorch, NumPy, Scanpy, Seurat, ggplot2; Linux, Git

Statistical Methods Bayesian inference, variational inference, integrative modeling with latent factors, causal mediation analysis, integrative clustering, longitudinal data analysis, survival analysis, missing data, multiple testing under dependence

Machine Learning Representation learning, self-supervised learning, computer vision, transformer-based models, transfer learning

Computational Biology Multi-omics integration, transcriptomics, cell imaging and morphology, statistical genetics

Computing GPU/CUDA, Unix & cluster computing environment, scientific software development

SELECTED PUBLICATIONS

Praxis-BGM: Clustering of Omics Data Using Semi-Supervised Transfer Learning for Gaussian Mixture Models via Natural-Gradient Variational Inference.

Jia Q, Goodrich JA, et al.

Bioinformatics, 2026.

A False Discovery Rate Control Method Using a Fully Connected Hidden Markov Random Field for Neuroimaging Data.

Kim T, Jia Q, de Leon MJ, Shu H; Alzheimer's Disease Neuroimaging Initiative.

Medical Image Analysis, 2026.

LUCIDus: An R Package for Implementing Latent Unknown Clustering by Integrating Multi-omics Data With Phenotypic Traits.

Zhao Y, Jia Q, Goodrich JA, Conti DV.

The R Journal, 2024.

Integrating Multi-Omics with Environmental Data for Precision Health: A Novel Analytic Framework and Case Study on Prenatal Mercury-Induced Childhood Fatty Liver Disease.

Goodrich JA, Wang H, Jia Q, et al.

Environment International, 2024.

An Extension of Latent Unknown Clustering Integrating Multi-omics Data (LUCID) Incorporating Incomplete Omics Data.

Zhao Y, Jia Q, Goodrich JA, Darst B, Conti DV.

Bioinformatics Advances, 2024.

DeepFDR: A Deep Learning-Based False Discovery Rate Control Method for Neuroimaging Data.

Kim T, Shu H, Jia Q, de Leon MJ, et al.

Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR, 2024.

BiTr-UNet: A CNN-Transformer Combined Network for MRI Brain Tumor Segmentation.

Jia Q, Shu H.

MICCAI BrainLes Workshop, 2021.

PRESENTATIONS

Praxis-BGM: A Prior-Augmented and Regularized Natural-Gradient Variational Inference on Gaussian Mixture Models

Joint Statistical Meetings (JSM), Nashville, TN, Aug. 2025

BiTr-UNet: A CNN-Transformer Combined Network for MRI Brain Tumor Segmentation

MICCAI BrainLes Workshop, Virtual, Sep. 2021